

NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF COMMERCE

DEPARTMENT OF ACCOUNTING

MANAGEMENT AND COST ACCOUNTING II

PART 4.1

CAC 4105 MODULE

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Standard Costing and Variance Analysis

Introduction

The basic principles of standard costing and variance analysis are assumed to be familiar for all students at this stage. The initial content of this chapter amounts to a revision of the basic principles on the topic covered in both semesters of the second year. You are thus advised to devote adequate time to the revision of your second semester management accounting I and II. The PART 4.1 syllabus is cumulative and hence examination questions may draw heavily on material from second year studies.

Standard costing and variance analysis represent a particular approach to performance evaluation. The concept that underpins them is that efficiency can be monitored by periodically comparing actual costs incurred with standard costs for output achieved. This concept is not valid under all circumstances. In subsequent chapters, the module goes on to explore both the practice and limitations of standard costing.

Definitions

CIMA's Terminology defines standard costing as follows:

Standard costing: A control technique which compares standard costs and revenues with actual results to obtain variances which are used to stimulate improved performance.

The definition points to a very close relationship between standard costing and budgetary control (the practice of making continuous comparison between budget and actual results). They both compare the actual results with the expected performance to identify any variances. The difference is that with standard costing the comparison is usually made at a unit level, that is, the actual cost per unit is compared with the standard cost per unit. The resulting variances may be analysed to show their causes and we will see how this is done later in this chapter.

What is a standard Cost?

A standard cost is an estimated or predetermined cost of performing an operation or producing a good or service under normal conditions. This is a cost that should be incurred under efficient operating conditions.

A standard cost is not the same as a budgeted cost. A budget relates to an entire activity or operation; a standard presents the same information on a per unit basis. A standard therefore provides cost expectations per unit of activity and a budget provides the cost expectation for the total activity.

The standard cost may be stored on a standard cost card. The standard cost may be prepared using either absorption costing principles or marginal costing principles. The example which follows is based on absorption costing. Standard costs and standard prices provide the basic unit information which is needed for valuing budgets and for determining total expenditures and revenues.

A Standard

It is defined as a benchmark measurement of resource usage, set in defined conditions. The definition goes on to describe a number of bases which can be used to set the standard, including:

a prior period level of performance by the same organisation

the level of performance required to meet organisational objectives.

Use of the first basis indicates that management feels that performance levels in a prior period have been acceptable. They will then use this performance level as a target and control level for the forthcoming period. When using the second basis management is being more outward looking, perhaps attempting to monitor their organisation's performance against 'the best of the rest'. The third basis sets a performance level which will be sufficient to achieve the objectives which the organisation has set for itself.

Ideal standard

Standards may be set at ideal levels, which make no allowance for normal losses, waste and machine downtime. This type of ideal standard can be used if managers wish to highlight and monitor the full cost of factors such as waste, etc., however, this type of standard will almost always result in adverse variances since a certain amount of waste, etc., is usually unavoidable. This can be very demotivating for individuals who feel that an adverse variance suggests that have performed badly.

Attainable standard

Standards may also be set at attainable levels which assume efficient levels of operation, but which include allowances for factors such as normal loss, waste and machine downtime. This type of standard does not have the negative motivational impact that can arise with an ideal standard because it makes some allowance for unavoidable inefficiencies. Adverse variances will reveal whether inefficiencies have exceeded this unavoidable amount.

Basic Standard

A basic standard is one which is kept unchanged over a period of time. It is used as the basis for preparing more up-to-date standards for control purposes. A basic standard may be used to show the trend in costs over a period of time.

Standard costing in the modern industrial environment

There has recently been some criticism of the appropriateness of standard costing in the modern industrial environment. The main criticisms include the following:

(a) Standard costing was developed when the business environment was more stable and operating conditions were less prone to change. In the present dynamic environment, such stable conditions cannot be assumed.
(b) Performance to standard used to be judged as satisfactory, but in today's climate constant improvement must be aimed for in order to remain competitive.
(c) The emphasis on labour variances is no longer appropriate with the increasing use of automated production methods.

An organisation's decision to use standard costing depends on its effectiveness in helping managers to make the correct decisions. Standard costing may still be useful even where the final output is not standardised. It may be possible to identify a number of standard components and activities for which standards may be set and used effectively for planning and control purposes. In addition, the use of demanding performance levels in standard costs may help to encourage continuous improvement.

What is variance analysis?

A variance is the difference between the expected standard cost and the actual cost incurred. A unit standard cost contains detail concerning both the usage of resources and the price to be paid for the resources. Variance analysis involves breaking down the total variance to explain how much of it is caused by the usage of resources being different from the standard, and how much of it is caused by the price of resources being different from the standard.

These variances can be combined to reconcile the total cost difference revealed by the comparison of the actual and standard cost. A variance is said to be favourable if it causes actual profit to be greater than budget; it is said to be adverse if it causes actual profit to be less than budget.

We are going to look at each variance and discuss the meaning of each variance, so it is easy for you to logically deduce the formulae.

Total Direct materials variance

The direct material variance is the difference between the standard cost (SC) of materials resulting from production activities and the actual costs (AC) incurred.

$$SC - AC$$

This variance is further broken down into:

1. Direct material price variance
This is the difference between the standard number of units (SQ) and actual number of units (AQ) used in the production process, multiplied by the standard price per unit. This variance is the responsibility of the production department.

$$(SP - AP) \times AQ$$

2. Direct materials usage variance
This is the difference between the standard number of units (SQ) and actual number of units (AQ) used in the production process, multiplied by the standard price per unit. This variance is the responsibility of the production department.

$$(SQ - AQ) \times SP$$

Direct materials usage variance can be further broken down into:

2.1 Raw materials mix variance
This variance should be calculated when the units being produced have two or more raw materials that need to be mixed in certain proportions. It arises as a result of a change in the mix/proportions of raw materials in the actual production to the mix/proportions of raw materials in standard production. It is calculated as follows:

	Actual Mix	Standard mix in Actual Output.	Variance (units)	Standard Cost per Unit/kg	Variance $
Direct material X	xxx	xxx	xxx	xxx	xxx
Direct material Y	xxx	xxx	xxx	xxx	xxx
Direct material Z	xxx	xxx	xxx	xxx	xxx
	xxx	xxx			xxx

2.2 Raw materials yield variance

This variance arises as a difference between the standard/expected yield and the actual yield which has been obtained. This variance is "output based" as opposed to the materials mix variance, which is "input based". It is calculated as follows:

Actual Yield				xxx
less: Expected yield				xxx
Variance (units)				xxx
x direct material price/unit				xx
Variance (In $)				xxx

Example 1:

GRV is a chemical processing company that produces sprays used by farmers to protect their crops. One of these sprays is made by mixing three chemicals. The standard material cost details for 1 litre of this spray is as follows:

	$
0.4 litres of chemical A @ $30 per litre	12.00
0.3 litres of chemical B @ $20 per litre	6.00
0.5 litres of chemical C @ $15 per litre	7.50
Standard material cost of 1 litre of spray	25.50

During August GRV produced 1,000 litres of this spray using the following chemicals:

600 litres of chemical A costing $18,000
250 litres of chemical B costing $8,000
500 litres of chemical C costing $8,500

Required: Calculate the Material Mix and yield variances.

Material Mix Variance

	Actual Input in Actual Mix	Standard mix in Actual Output.	Variance	Standard Cost per litre in $	Variance $
Chemical A	600	450	150A	30	4 500 A
Chemical B	250	337.5	87.5F	20	1 750 F
Chemical C	500	562.5	62.5F	15	937.5 F
	1350	1350			1 812.5 F

Material Yield Variance

Actual Yield	1000
less: Expected yield (1350 / 1.2)	1125
Variance (units)	125 Adverse
x direct material price/unit {(0.4 x 30)+ (0.3 x 20)+ (0.5*15)}	25.5
YieldVariance (In $)	3 187 . 5 (A)
Actual usage (in kgs)	1 350
less: Expected usage (1000*0.4+1000*0.3+0.5*1000)	1 200
Variance (units)	150A
x weighted material price/litre 25.5/ 1.2	21.25
Yield Variance (In $)	3 187 .50 (A)

Total Direct labour variance

The total direct labour variance is the difference between the standard labour cost (SC) for the actual production and the actual labour cost (AC).

$$SC – AC$$

This variance is further broken down into:

1. **Direct labour rate variance**

The labour rate variance measures the difference between the actual labour rate paid (AR) and the standard rate (SR), multiplied by the number of actual hours worked (AH).

$$(SR – AR) \times AH$$

2. **Direct labour efficiency variance**

The labour efficiency variance measures the ability to utilize labour in accordance with expectations. The variance is useful for spotlighting those areas in the production process that are using more labour hours than anticipated. This variance is calculated as the difference between the actual labour hours (AH) used to produce an item and the standard labour hours (SH) that should have been used, multiplied by the standard labour rate (SR).

$$(SH – AH) \times SR$$

Note	*At this level, a distinction has to be made between* ***labour hours worked*** *and* ***labour hours paid.*** *When calculating the labour efficiency variance* ***labour hours worked*** *should be used whereas when calculating the Labour rate variance,* ***labour hours paid should be paid.***

3. **Labour Idle time variance**

It indicates the difference which exists between actual hours spent in active production viz a viz actual labour hours paid. In a normal/realistic production scenario the production process is affected by factors beyond the control of the management, e.g. machinery breaks down. During these times production is halted but organisations have to pay full wages in full as per the labour act. The variance is calculated by the formulae:

(Actual Hours Paid – Actual hours worked) **x** Standard labour rate

Total variable overhead variance

A total variable overhead variance is calculated in the same way as the total direct material and labour variances. The total variable overhead variance is the difference between standard variable overheads charged to production (SC) and actual variable overheads incurred (AC).

SC – AC

This variance is further broken down into:

1. Variable overhead expenditure variance

To compare the actual overhead expenditure with the budgeted expenditure, we need to "flex the budget" first. This process means changing a budget to allow for different output levels and allows more realistic variance analysis at the end of the financial period in question. The variable overhead expenditure variance is the difference between the budgeted flexed variable overheads (BFVO) and actual variable overhead (AVO).

BFVO – AVO

Or

(SVOAR – AVOAR) AH

SVOAR – Standard Variable Overhead Absorption Rate

AVOAR – Actual Variable Overhead Absorption

2. Variable overhead efficiency variance

This is the difference between standard hours of output (SH) and actual hours of input (AH) for the period multiplied by the standard variable overhead rate (SVOAR).

(SH – AH) × VOAR

Total fixed overhead variance

Fixed overhead total variance is the difference between actual (AC) and absorbed fixed production overheads (SC) during a period. This applies where an absorption costing system has been implemented in order to determine the foxed overhead absorption rate.

SC – AC

Since we are using an absorption costing system, an under- or over-recovery of overheads may arise if actual output or overhead expenditure differs from budget. The total fixed overhead variance is broken into:

1. Fixed overhead expenditure variance

This is the difference between the budgeted fixed production overhead (BFO) and the actual fixed production overhead (AFO).

BFO – AFO

2. Volume variance

This variance arises as a result of the difference between actual production (AP) and budgeted production (BP), multiplied by the standard fixed overhead absorption rate (FOAR).

Volume = (Budgeted production – Actual Production) **x** FOAR

The volume variance reflects that fixed costs do not change with changes in production in the short term. The volume variance is further divided into:

2.1 Fixed Overhead Capacity Variance
This variance arises as a result of the difference Budgeted Production Hours (BH) and Actual hours (AH), multiplied by the Standard Fixed overhead absorption rate (FOAR):

(BH – AH) **x** FOAR

2.2 Fixed Overhead Efficiency variance
This variance arises as a result of the difference Standard Production Hours (SH) and Actual hours (AH), multiplied by the Standard Fixed overhead absorption rate (FOAR):

(SH – AH) **x** FOAR

Sales variance

These variances are used to analyse the performance of the sales function or revenue centres. Total sales variance is calculated as the difference between actual sales revenue (ASR) less standard variable cost of sales (SCOS) and budgeted contribution (BC).

(ASR – SCOS) – BC

Sales Price Variance

This is the difference between the actual selling price (ASP) and the standard selling price (SSP) multiplied by the actual sales volume/quantity.

(ASP – SSP) × AV

Sales volume variance

This is the difference between the actual sales volume (AV) and the budgeted volume (BV) multiplied by the standard contribution margin (SM).

(AV – BV) × SM

Sales volume can be further broken down into:

a. Sales margin mix variance
This variance arises when there is a change between the expected mix of sales quantities and the actual mix of sales quantities where an organisation sells more than one product. It is the difference between the actual volume in budgeted proportions (AVBP) and actual volume sold (AV) multiplied by standard margin (SM).

(AVBP – AV) × SM

b. Sales margin quantity variance
This variance seeks to identify the difference between the budgeted and actual sales volumes (holding the product mix constant). It is therefore the difference between the

actual sales volume in budgeted proportion (AVBP) and the budgeted sales volume (BSV) multiplied by the standard margin (SM).

$$(AVBP – BSV) \times SM$$

Practice Questions

Question 1 - MATOJENI LTD

Month 1

Matojeni Ltd processes and sells brown rice. It buys unprocessed rice seeds and then, using a relatively simple process, removes the outer husk of the rice to produce the brown rice. This means that there is substantial loss of weight in the process. The market for the purchase of seeds and the sales of brown rice has been, and is expected to be, stable. Matojeni Ltd uses a variance analysis system to monitor its performance.

There has been some concern about the interpretation of the variances that have been calculated in month 1.

1. The purchasing manager is adamant, despite criticism from the production director, that he has purchased wisely and saved the company thousands of dollars in purchase costs by buying the required quantity of cheaper seeds from a new supplier.
2. The production director is upset at being criticised for increasing the wage rates for month 1; he feels the decision was the right one, considering all the implications of the increase. Morale was poor and he felt he had to do something about it.
3. The maintenance manager feels that saving $8 000 on fixed overhead has helped the profitability of the business. He argues that the machines' annual maintenance can wait for another month without a problem as the machines have been running well.

The variances for month 1 are as follows:

		$
Material price	Favourable	48 000
Material usage	Adverse	52 000
Labour rate	Adverse	15 000
Labour efficiency	Favourable	18 000
Labour idle time	Adverse	12 000
Variable overhead expenditure	Adverse	18 000
Variable overhead efficiency	Favourable	30 000
Fixed overhead expenditure	Favourable	8 000
Sales price	Adverse	85 000
Sales volume	Adverse	21 000

Matojeni Ltd uses labour hours to absorb the variable overhead

Month 2

In month 2 the following standard data applies:

Standard costs for 1 tonne of brown rice:

- 1,4 tonnes of rice seeds are needed at a cost of $60 per tonne
- It takes 2 labour hours of work to produce 1 tonne of brown rice and labour is normally paid $20 per hour.
- 2 hours of variable overhead at a cost of $30 per hour
- The standard selling price is $240 per tonne
- The standard contribution is $56 per tonne

Budget information for month 2 is:

- **Fixed costs were budgeted at $210 000 for the month**
- **Budgeted production and sales were 8 400 tonnes**

The actual results for month 2 were as follows:

- Actual production and sales were 8 000 tonnes
- 12 000 tonnes of rice seeds were bought and used costing $660 000
- 15 800 labour hours were paid for costing $303 360
- 15 000 labour hours were worked
- Variable production overhead cost $480 000
- Fixed costs were $200 000
- Sales revenue achieved was $1 800 000

REQUIRED		**Marks**	
		Sub-total	**Total**
(a)	Using the variances and other relevant information above comment on the performance of the following managers in **month 1:** (i) The purchasing manager (ii) The production director (iii) The maintenance manager	(10)	(10)
	Note : You are not required to comment on the performance of the business or its managers in month 2.		
(b)	Calculate the variances for month 2 in as much detail as the information above allows.	10	10
(c)	Reconcile the budget profit to the actual profit using marginal costing principles.	(15)	(15)
Total for Practice Question 1			**35**

Solution to Practice Question 1.

(a) Using the variances and other relevant information above comment on the performance of the following managers:

(i) The purchasing manager

- The purchasing manager has clearly bought a cheaper product, saving $48 000. The cause of this is not specified and it could be due to good buying or negotiation, reductions in quality or changes in overall market conditions.
- We are told the market for buying seeds is stable, so there is more likely to be an internal reason for the problem. The material usage variance is significantly adverse, indicating that much more waste than normal has occurred in month 1.
- This suggests that the quality of the seed bought was poor and as a result a $52 000 excess loss has occurred. It is possible that the waste was caused by the labour force working poorly or too quickly and this has to be considered.
- The sales price achieved is also well down on standard with the sales price variance showing an $85 000 loss of revenue and (therefore) profit.
- We are told that the market for sales of brown rice is stable and so it is reasonable to presume that the fall in sales price achieved is as a result of internal quality issues rather than general price falls. This may have also led to a fall in the volume of sales, another $21 000 of adverse variance.
- In conclusion the purchasing manager appears mainly responsible for a loss of $110 000 ($85 000 + $52 000 + $21 000 – $48 000) taking the four variances above together.

(ii) The production director.

- The production director has increased wage rates and this has cost an extra $15 000 in month 1. However, one could argue that this wage increase has had a motivational effect on the labour force.
- The labour efficiency variance is $18 000 favourable; and so it is possible that a wage rise has encouraged the labour force to work harder. Academic evidence suggests that this effect might only be temporary as workers get used to the new level of wages.
- The plant has recorded an adverse idle time variance of $12 000. Idle time can have many causes, including, material shortages, machine breakdowns, unclear work instructions and poor bill of materials/assembly drawings. However, we are told the machines are running well and the buyer has bought enough rice seeds. This could point to a probably once – off incidence of unclear work instructions or poor bill of materials/assembly drawings as a probable cause.
- In conclusion the increase in the wage rate did cost more money despite a marginal improvement in morale and productivity. The total of the three variances above is $9 000 Adverse ($18 000 – $12 000 – $15 000).

(iii) The maintenance manager.

- The maintenance manager has decided to delay the annual maintenance of the machines and this has saved $8 000. This will increase profits in the short term but could have disastrous consequences later.
- In this case only time will tell. If the machines breakdown before the next maintenance, then lost production and sales could result.

- • The maintenance manager has only delayed the expenditure and not prevented it altogether. A saving of $8 000 as suggested by the variance has not been made. It is also possible that the adverse variable overhead expenditure variance has been at least partly caused by poor machine maintenance.
- The variance calculated is not the saving made as it represents a timing difference only. The calculation also ignores the risks involved.

(b) Calculate the variances for month 2 in as much detail as the information above allows

 The status of a variance (adverse or favourable) should be clearly indicated. Where this is not done or incorrectly done, the deduction is that the student does not understand the topic or underlying principle. Note that the sales volume variance is expressed in terms of contribution and that the FOH variance in this case only referred to the expenditure variance.

Sales Price Variance

Formulae	(SP – AP) AQ
Variance [AP: $1 800 000 / 8 000 tonnes = $225 per tonne]	($240 – $225)8 000 t
	$120 000 Adverse (1)

Sales volume

Formulae	(BQ – AQ) Std Contribution/Unit
Variance	(8 400 t – 8 000 t) $56
	$22 400 Adverse

Direct material price

Formulae	(SP – AP) AQ
Variance [AP: R660 000 / 12 000 tonnes = R55 per tonne]	($60 – $55)12 000 t
	$60 000 Favourable

Direct material usage

Formulae	(SQ – AQ) SP
Variance [SQ: 8 000 tonnes rice x 1,4 = 11 200 tonnes seed]	(11 200 t – 12 000 t) $60
	$48 000 Adverse

Direct labour rate

Formulae	(SR – AR) AH
Variance [AR: $303 360 / 15 800 hrs = $19,20 per hour]	($20 – $19,20)15 800 hrs **$12 640 Favourable**

Labour idle time

Formulae	(AHP – AHW) SR	
Variance	(15 800 hrs – 15 000 hrs) R20 **$16 000 Adverse**	(1)

Direct labour efficiency

Formulae	(SHA – AHW) SR
Variance [SHA: 8 000 tonnes x 2 hrs = 16 000 hrs]	(16 000 hrs – 15 000 hrs) $20 **$20 000 Favourable**

Direct labour efficiency

Formulae	(SHA – AHW) SR
Variance [SHA: 8 000 tonnes x 2 hrs = 16 000 hrs]	(16 000 hrs – 15 000 hrs) $20 **$20 000 Favourable**

Variable overhead expenditure

Formulae	(SR – AR) AHW
Variance [AR: $480 000 / 15 000 hrs = $32 per hr]	($30 – $32)15 000 hrs **$30 000 Adverse**

Variable overhead efficiency

Formulae	(SHA – AHW) SR
Variance [SHA: 8 000 tones x 2 hrs = 16 000 hrs]	(16 000 hrs – 15 000 hrs) $30 **$30 000 Favourable**

Fixed overhead expenditure

Formulae	Budget – Actual
Variance	$210 000 – $200 000 **$10 000 Favourable** (1)

Part C Reconcile the budget profit to the actual profit using marginal costing principles.

The question requires the student to prepare an operating statement using Marginal/direct costing principles. Take note of the required formats. Students will not be awarded any marks at this stage if a different format is used.

			$
Budgeted Contribution ($ 56 **x** 8 400t)			470400
Sales volume contribution variance			-22400
Flexible Budget Contribution (8000 **x** 56**)**			448000
Selling price varaiance			-120000
			328000
Vaiable Cost variances	Favourable	Adverse	
	$	$	
Material Price	60000		
Material usage		-48000	
labour rate	12640		
labour idle time		-16000	
Labour efficiency	20000		
Vaiable overhead expenditure		-30000	
Variable overhead efficiency	30000		
Total	122640	-94000	28640
Actual contribution			356640
Budgeted fixed production overhead			210000
Fixed overhead expenditure variance			-10000
Actual profit			156640

Practice Question 2 - adopted from Nust CAC 2205 – 2022 question paper.

Question 3

Malcom Reynolds Ltd makes and sells a single product, Product Q, with the following standard specifications for materials:

	Quantity	Price per kg
	Kg	$
Direct material X	12	40
Direct material Y	8	32

It takes 20 direct labour hours to produce one unit with a standard direct labour cost of $10 per hour. The annual sales/production budget is 2 400 units evenly spread throughout the year. The standard selling price was $1 250 per unit. The budgeted production overhead, all fixed is $288 000 and expenditure is expected to occur evenly over the year. Absorption is based on direct labour hours. For the month of October, the following actual information is provided.

	$	$
Sales (220 units)		264 000
Cost of sales		
direct materials	159 000	
direct wages	45 400	
Fixed production overhead	23 000	227 400
Gross Profit		36 600
Administration costs	13 000	
Selling and distribution costs	8 000	21 000
Net Profit		15 600

Cost of opening stocks, for each material, were at the same price per kilogram as the purchases made during the month but there had been changes in the materials stock levels,

	1-Oct	30-Oct
	Kgs	Kgs
Material X	680	1 180
Material Y	450	350

The number of direct labour hours worked was 4 600 hours and the total wages incurred $45 400. Work in progress stocks and finished maybe assumed to be the same at the beginning and end of October.

Required:

a) Calculate the standard product cost for one unit of product Q showing the standard selling price and standard gross profit margin. **[5]**

b) Calculate appropriate variances for the materials, labour, fixed production overhead and sales, noting that it is company policy to calculate material price variances at the time of issue to production**.** **[15]**

c) Prepare a statement for management reconciling the budgeted gross profit with the actual gross profit**.** **[5]**

Solution to practice question 2

(a) Detemination of Standard Product Costs

		$	$
Selling price			1250
less: **Production Costs**			
Direct material X	12 x 40	480	
Direct material Y	8 x 32	256	
Direct labour	20 x 10	200	
Fixed production overhead	(288000/24000) = $6/hour x 20 hours	120	1056
Gross Profit / unit			194

(b) Calculation of variances

Material Price Variance

Formulae	**(SP - AP) AQ**	
Material X	(40 - 42) 2500 (w1)	**$ 5 000 Adverse**
Material Y	(32 - 30) 1800 (w1)	**$ 3 600 Favourable**

Material Usage Variance

Formulae	**(SQ - AQ) SP**	
Material X	(2 640 - 2 500) 40	**$ 5 600 Favourable**
Material Y	(1 760 - 1 800) 32	**$ 1 280 Adverse**
Total material Usage variance		**$ 4 320 Favourable**

Material Mix Variance

	Actual Mix	**Standard mix in Actual Output.**	**Variance**	**Standard Cost per Unit/kg in $**	**Variance $**
Direct material X	2 500	2 580	80 F	40	3 200 F
Direct material Y	1 800	1 720	80 A	32	2 560 A
	4 300	4 300			640 F

Material Yield Variance

Actual Yield	220
less: Expected yield (4300 / 20)	215
Variance (units)	5 Favourable
x direct material price/unit {(12 x 40)+ (8 x 32)}	726
Yield Variance (In $)	3 680 (F)

Or

Actual usage (in kgs)	4 300
less: Expected usage (220*12+220*8)	4 400
Variance (units)	100F
x weighted material price/kg {(12 x 40)+ (8 x 32)}/ (8+12)	36.8
Yield Variance (In $)	3 680 (F)

Labour rate variance

Formulae **(SR - AR) AH**

(10 - 9.87) 4 600

$600 - 00 (F)

Note	*The majority of students made the mistake of rounding off the actual labour rate. Calculations made before obtaining the final solution should be stored in the calculator. Only the final solution should be rounded off. Answers like $598 although very close to the final solution where not marked.*

Labour efficiency variance

Formulae **(SH - AH) SR**

(4 400 - 4 600) 10

$2 000 (A)

Fixed Production Variances

Expenditure

Formulae **Budgeted - Actual**

288 000/12 - 23 000

24 000 - 23 000

$ 1 000 (F)

Volume

Formulae **(Budgeted Pdn - Actual Pdn) FOAR**

(200 - 220) 120

$2 400 - 00 (F)

Capacity

Formulae **(BH - AH) FOAR**

(4 000 - 4 600) 6

$3 600 - 00 (F)

Efficiency

Formulae **(SH - AH) FOAR**

(4 400 - 4 600) 6

$1 200 - 00 (A)

Sales price Variance

Formulae **(SSP - ASP) ASQ**

(1 250 - 1 200) 220

$1 100 - 00 (A)

Sales Volume Variance

Formulae **(SSQ - ASQ) Standard Profit**

(200 - 220) 194

$3 880 - 00 (F)

Operating statement under absorption costing.

				$
Budgeted profit				38 800
Sales volume profit variance				3 880
Flexed budget profit				42 680
Selling price variance				(11 000)
				31680
Cost variances		**Favourable**	**Adverse**	
		$	$	
Material Price - Material X			(5 000)	
- Material Y		3 600		
Material Usage Material X		5 600		
Material Y			(1 280)	
Labour Rate		600		
Labour efficeincy			(2 000)	
Fixed Production Expenditure		1 000		
Fixed Production Capacity		3 600		
Fixed production Efficiency			(1 200)	
Total		14400	-10480	3920
Actual Profit				35600

Nust Past Exam Paper Question – 2021 CAC 4105

QUESTION 3 – STANDARD COSTING

Bela-Bela Cleaning Products (Pvt) Ltd ('BBCP') is a company situated in Dulibadzimu, a town in Beitbridge, and manufactures and distributes various cleaning products.

BBCP Soap Manufacturing Division

This division manufactures and supplies environmentally-friendly bars of soap to leisure, nature and wildlife lodges for use in their chalets and various facilities. The bars weigh 100g each and are sold for $5,50 each. They are sold in bags containing 10 bars of soap.

The Soap Manufacturing Division of BBCP manufactures the soap using an entirely automated production process. Supervision and maintenance costs fall under overhead.

The following information regarding the cost of a bag of 10 soap bars is available:

Input at standard cost:

0,30kg	of Ingredient C	at $38,00 per kg
0,65kg	of ingredient T	at $13,00 per kg
0,2kg	of Ingredient A	at $21,00 per kg
1,15kg		

Other information:

Variable overhead: 0,25 machine hours
Fixed overhead: 0,25 machine hours
Packaging: one (1) plastic bag at $0,30 per bag

Overheads are absorbed on the basis of machine hours. BBCP uses an absorption costing system.

The following budget information is available regarding total overhead costs (i.e. total variable overhead cost plus total fixed overhead cost) at different production levels (based on variable costing principles):

Month	Production Level Bags of Soap	Total overhead cost (fixed + variable) $
1	2 320	30 220
2	2 500	31 075
3	2 200	29 650

The January 2022 budget includes 600 machine hours and the production of 2 400 bags of soap, which was the most common production level of the past two years.

The actual results for January 2022 show the following:

- 2 490 bags of soap were produced.
- 750 kg of ingredient C was purchased at $39,00 per kg. 730 kg was issued to production.
- 1 550 kg of ingredient T was purchased for $20 460. 1 600 kg was issued to production.
- 560 kg of ingredient A was purchased at $18,50 per kg. 550 kg was issued to production.
- Variable overhead cost of $12 864 was incurred for 640 machine hours.
- The actual fixed overhead incurred was $20 000.
- Plastic bags were purchased at $0,30 per bag. The total cost amounted to $747

Inventory of raw materials, packaging and finished goods are valued at standard cost. The standard costs have not been changed for the past six months of 2021.

QUESTION 3			
REQUIRED		**Marks**	
		Sub-total	**Total**
(a)	Calculate the standard cost of one bag of soap bars for Bela-Bela Cleaning Products (Pty) Ltd (BBCP) using an absorption costing basis.	(5)	(5)
(b)	Calculate the following standard costing variances for BBCP's soap division for the month of January 2022: • Material price variances for ingredients C, T and A. • Material mix and yield variances for ingredients C, T and A. • Variable overhead expenditure variance and efficiency variance. • Fixed overhead expenditure variance and volume variance.	(3) (8) (2) (2)	(15)
Total for Question 3			**20**

(a) Calculate the standard cost of one bag of soap bars for Bela-Bela Cleaning Products (Pvt) Ltd using an absorption costing basis.

High-low method:

Variable cost per unit = $\frac{\text{Cost at highest activity level – Cost at lowest activity level}}{\text{Units produced at highest activity level – units produced at lowest activity level}}$

Variable cost per unit = $\frac{\$31\,075 - \$29\,650}{2\,500 - 2\,200}$ (1) r/w (1) r/w

= \$1 425 / 300 = \$4,75 per bag

[\$4,75 / 0,25 = \$19 per hour]G3

Total variable cost for 2 500 units = 2 500 x \$4,75 = \$11 875 (½) c
Therefore fixed cost = \$31 075 – \$11 875 = \$19 200 per month (½) c

Fixed overhead rate = \$19 200 / 600 hrs = \$32,00 per hour

Standard cost for 1 bag (1kg) (i.e. 100g x 10 = 1kg)

		$	
Ingredient C	0,30kg x $38,00 /kg	11,40	(½) r/w
Ingredient T	0,65kg x $13,00 /kg	8,45	(½) r/w
Ingredient A	0,20kg x $21,00 /kg	4,20	(½) r/w
		24,05	
Variable overhead	(0,25 hr x $19,00 /hr)	4,75	(½) r/w
Fixed overhead	0,25 hr x $32,00 /hr	8,00	(½) r/w
Packaging	1 x $0,30	0,30	(½) r/w
		37,10	
	Available		**6**
	Maximum		**5**

Part (b) Calculate the following standard costing variances for BBCP 's soap division for the month of August 2015: **Note: the final variance amounts must be in Rand value**

Material price variances

		$	
Ingredient C	750kg x ($39 – $38)	(750)A	(1) r/w
Ingredient T	1 550kg x ($20 460/ 1550= $13,20 – $13)	(310)A	(1) r/w
Ingredient A	560kg x ($18,50 – $21)	1 400 F	(1) r/w
		340 F	

Yield Variance

Actual input	2 880kg
Expected output (2 880 x 1/1,15)	2 504,348 kg
Actual output	2 490kg
Yield variance kg(negative) (2 490kg - 2 504,345kg)	14,348 kg
Yield variance ($) (14,345kg x $24,05 /kg)	$345 A

Material mix variances	**Actual input in actual mix (kg)**	**Actual input in std mix (kg)**	**Variance (kg)**	**Std price $**	**Variance $**	(1) r/w
Ingredient C (2 880 x (0,3 / 1,15))	730	751,3	21,3	38,00	**809,40 F**	
Ingredient T (2 880 x (0,65 / 1,15))	1 600	1 627,8	27,8	13,00	**361,40 F**	(1) r/w
Ingredient A (2 880 x (0,2 /1,15))	550	500,9	(49,1)	21,00	**(1 031,10)A**	(1) r/w
	2 880	2 880,0			139,70 F	

Variable overhead expenditure variance

[$4,75 / 0,25 = $19 per hour]
= $12 864 – (640 hrs x $19) (1) r/w
= $12 864 – $12 160
= $704 Adverse

Variable overhead efficiency variance

= [(2 490kg x 0,25 hr/kg) – 640hr] x $19 /hr (1) r/w
= (622,5 hr – 640 hr) x $19 /hr
= -17,5 hr x $19 /hr
= $332,5 Adverse

Fixed overhead expenditure variance

= $19 200 – $20 000 (1) r/w
= $800 Adverse

Fixed overhead volume variance

= (2 490kg x $8) – $19 200 OR [(2 490 – 2 400) x $8]
= $19 920 – $19 200
= $720 Favourable (1) r/w

Communication skills mark: Variance type clearly indicated (1)

Available and maximum 15

Practice Question 4

K&S LTD

K & S Ltd manufactures floor polish and liquid soaps for commercial clients. Both product lines are sold in 10 litre containers. The company is critically aware of their trading environment and uses a standard costing system, which is reviewed every six months. The last review was done on 1 March 2013 for the six months ending 28 August 2013.

Floor polish division

The following standard cost was set for one 10-litre container of polish:

	Standard Cost	Unit Cost
	$	$
Direct material		
• 5 litre policloro	$3.60/litre	18
• 3.5 litre oxotone	$2.20/litre	7.7
• 4 litre lather	$0.6/litre	2.4
Direct labour: 0,24 productive hours	$21.875/hr	5.25
Variable overhead: 0,24 hours		14.88
Fixed overhead: 0,24 hours		8.4
Container		5.1
Total standard cost per unit		61.73
Sales price per unit		90

Budgeted idle time of 4% was considered in detailing the above standard cost card. K & S allocates both variable and fixed overhead costs to production on the basis of productive labour hours. The budgeted fixed manufacturing overhead for the six months is $10 320 000. Manufacturing activities and incurring of production cost take place evenly throughout a year.

The actual costs incurred during March 2013 amounted to:

Purchase of direct materials:

Policloro	1 100 000 litres	@ $3,80 / litre
Oxotone	725 000 litres	@ $2,30 / litre
Lather	830000 litres	@$0,56 / litre
Labour	53800 clocked hours @ $21.50/hour(49 920) worked hours	
Variable Overhead	$3 072 800	
Fixed Overhead	$1 689 200	
Containers:	212 000 containers @ $5,10 each	

The actual production for March 2013 was 210 000 units. All containers purchased were issued. 1 000 000 litres of Policloro, 790 000 litres of Oxotone and 850 000 litres of Lather were issued to production.

REQUIRED		**Marks**
(a)	For the Floor polish division, calculate the Material Labour Variances for March 2013 in as much detail as possible.	(8)
Total		8

Practice Question 5 – Sales mix and quantity variances

Famous Fishing West Coast (Pty) Ltd (FFWC), operates a fleet of specialised fishing vessels that catch pilchards.

In catching Pilchards FFWC is governed to limit the quantity of each species of fish that may be caught each year. This is called the Total Allowable Catch (TAC) for the year.

The TAC for pilchards has been declining over the past three years. The local and African market for canned pilchards has, however, continued to grow. FFWC has therefore experienced a shortage of pilchards to supply the market and maintain its market share. In the past FFWC overcame the shortage by importing canned pilchards from Thailand and putting their own label on the tins. This left the cannery section of their Saldanha factory with spare capacity and seasonal cannery workers working less hours. FFWC sells all its pilchards in 400g cans. The decline in the Rand and subsequent Rand volatility presented challenges for FFWC. During the financial year ended 31 August 2017, FFWC decided to change its strategy by reducing the import of canned pilchards and instead importing frozen pilchards. The frozen pilchards were then processed and canned in FFWC's own cannery in Saldanha Bay.

The **budgeted** sales quantity and production costs per ton for each type of pilchard canned for the year ended 31 August 2017 were as follows:

	Sales quantity (tonnes)	**Production cost per tonne**
Own catch	3 600	$14 260
Imported cans	3 500	$17 200
Imported frozen pilchards	2700	$16 000

The budgeted average selling price for the year was $11,40 per 400g can of pilchards.

FFWC uses a standard absorption costing system. The budget was based on the standard revenues and costs.

The **actual** results for the year ended 31 August 2017 were as follows:

	Sales quantity (tonnes)	**Production cost per tonne**
Own catch	1 100	$13 900
Imported cans	3 400	$17 550
Imported frozen pilchards	4 800	$16 200

The actual average selling price for the year was $11,68 per 400g can of pilchards.

According to Stats SA, the compound annual growth rate in the retail price of canned pilchards since 2008 has been 6% which is more or less in line with inflation.

FFWC does not keep any inventory of pilchards at the beginning or end of the financial year (budget and actual).

QUESTION 1			
REQUIRED		**Marks**	
		Sub-total	**Total**
(a)	Calculate the following variances for FFWC's canned pilchards for the year ended 31 August 2017: (i) Sales price variance (ii) Sales margin mix variance (iii) Sales quantity variance *Communication skills – layout and structure*	4 6 4 *1*	15
(b)	Prepare a brief explanation of the reason for the difference between the budgeted and actual sales and operating profit figures for canned pilchards based on your calculations in part (a)		(10)
Total for Question 1			**25**

Practice Question 5

Air Woolies (Pty) Ltd (hereafter "the company" or "Air Woolies") is a local company which manufactures packaged foods and meals for supply to various airlines in South Africa as well as airlines based in other parts of Africa. The company recently entered into sales contracts with airlines in Botswana and Mauritius. The company claims to have "the tastiest food in thin air" and is looking to increase their product offering. One of their main products is packaged peanuts and raisins. This product is an age old favourite of flight passengers and Air Woolies has received large order quantities in recent months.

Albert Walton, the financial manager of Air Woolies, has told you that he is generally pleased with the company's financial results. However, he wants to understand why actual results varied from the budget. He approached you as a management accountant to look into this. There was no opening or closing stock of raw materials or finished goods.

The following Budget was prepared for sales of packets of peanuts and raisins for the month of November 2016:

	Notes	**ZWL Dollars**
Sales	1	225 000
Raw material: Peanuts	2	?
Raw material: Raisins	3	(10 800)

Packets	4	?
Direct labour	5	(25 000)
Variable overheads	6	(7 500)
Fixed overheads	7	(22 500)
Budgeted profit		?

Note 1:

The budgeted selling price per packet is $15. The actual selling price was $17 per packet and actual sales amounted to 12 800 packets for the month of November.

Note 2:

Each packet weighs 50 grams and should contain 55% peanuts and 45% raisins.

The company uses a specific type of peanut and has entered into a contract to import these peanuts from an overseas supplier at an agreed price of $4 per kilogram. The budget was based on an expected USD exchange rate of $1: ZWL$16. The budgeted cost of peanuts for November 2016 was USD$1 650.

Actual purchases and usage of peanuts was 385 kilograms.

The actual USD exchange rate for November was USD$1: ZWL$16.50

Note 3:

Raisins are purchased from a local supplier at a budgeted cost of $32 per kilogram. However, the supplier charged Air Woolies a discounted price of $30 per kilogram to retain their business.

Actual purchases and usage of raisins for the month of November was 280 kilograms.

Note 4:

Raisins and peanuts are packaged together in plastic packets which are specifically designed to be strong yet can also be opened easily by passengers. The budgeted price is $0.60 per packet with one packet being required for each peanut and raisin combination. Actual packet costs amounted to $0.80 per packet and 12 880 packets were purchased and used.

Note 5:

Labourers are required for packing the required quantities of ingredients in each packet. The budgeted labour rate is $25 per hour. It is expected that labourers will require 4 minutes to complete each packet. Actual labour hours worked amounted to 845 hours while actual labour costs were $23 660.

Note 6:

Allocation of variable overheads is linked to labour hours. Air Woolies budgeted on $7.50 of variable overheads for every labour hour worked. Actual variable overheads amounted to $5 070 for November 2016.

Note 7:

The company uses a standard absorption costing system.

Budgeted fixed overheads of $22 500 can be split between the following items:

- Rental
- Depreciation on machinery
- Electricity

Fixed overheads are allocated based on normal capacity of 15 000 packets per month. Actual fixed overheads were made up of the following:

- Rental $11 000
- Depreciation on machinery $6 000
- Electricity $8 000

REQUIRED:

a) Determine the flexed profit for November 2016. **(6)**

b) Perform a reconciliation of budgeted sales to actual profit for November 2016. Show all calculations and break variances down into as much detail as possible. Sales volume variance should be based on selling price. **(32)**

c) Provide reasonable explanations for all sales variances and all direct material variances calculated in part b above. **(8)**

d) Air Woolies is considering further expansion into Africa. Determine the strategic factors and additional considerations Air Woolies should take into account relating to their business of providing meals to airlines.

Suggested Solution

(a)

Budgeted sales = 225000/15 = 15000 units (packets)	**Budget**
Sales (given)	225 000
Raw Materials - Peanuts (16 x $1 650)	(26 400)
Raw materials - Raisins (given)	(10 800)
Packets (0.60 x 15 000)	(9 000)
Direct labour (given)	(25 000)
Variable overheads (given)	(7 500)
Fixed overheads (given)	(22 500)
Budgeted profit	**123 800**

To get to flexed = 123 800 x 12 800✓
/ 15 000 ✓ = 105 642.67✓**P**
Flexed profit

ALTERNATIVE:

Budgeted sales = 225 000/15 = 15 000
units (packets) Actual sales = 12 800
(given) Peanuts:
12 800 packets sold:
Each packet weighs 50 grams' peanuts required = 12 800 x 50 grams = 640 000/1 000 = 640 kilograms' x 55% = 352 kg of peanuts.
Raisins required = 640 kilograms' x 45% = 288 kg of raisins
Direct labour: 4 minutes per packet
12 800 x 4 = 51 200/60 = 853,33 hours

	Dollars	
Sales ($15 x 12 800)	192 000	✓
Raw materials - Peanuts ($4 x R16 x 352)	(22 528)	✓
Raw materials - Raisins ($32 x 288)	(9 216)	✓
Packets (0.60 x 12 800)	(7 680)	✓
Direct labour (853.33 x 25) **could use 854 (rounded up)**	(21 333) ✓	(rounded) or (21350)
Variable overheads (7.50 x 853.33) or (7.50 x 854) (allocated on actual hours)	(6 400 ✓	(rounded)) or (6 450)
Fixed overheads (use allocation rate: **22 500/ 15 000 x 12 800**)	(19 200)✓	
Flexed profit	**105 643✓P**	

Note: Flexed cost of sales= $192 000 – $105 643 = $86 357
Raw materials:

Peanuts:

Price variance:
Budgeted price = R16 x $4 = R64
Actual Price = R16.5 x $4 = 66
(AP-SP) x AQ
= (64-66) x 385kg = (770) A✓

Usage variance:

Total usage variance:
(AQ-SQ) x SP
(385-352) ✓ x 64 ✓= (2 112) A

Mix variance:
(AQ- AQ in standard mix) x SP
AQ in standard mix = (385 + 280) x 55% = 365.75

AQ = 385
(385 - 365.75) x 64= (1232) A✓
Yield variance:
(AQ in standard mix – SQ in standard mix) x SP
SQ in standard mix = 352 kg of peanuts
(365.75 - 352) x 64 = (880) A✓

Raisins:

Price variance:
(AP-SP) x AQ
(30-32) x 280kg = 560F✓

Usage variance:

Total usage variance:
(AQ-SQ) x SP
(280-288) ✓ x 32 ✓= 256F

Mix variance:
(AQ - AQ in standard mix) x SP
AQ in standard mix = (385 + 280) x 45%= 299.25
AQ = 280
(280 – 299.25) x 32 = 616F✓

Yield variance:
(AQ in standard mix – SQ in standard mix) x SP

SQ in standard mix = 288 kg of peanuts

(299.25 - 288) x 32 = (360) A ✓

Packets:

Price variance:
(AP - SP) x AQ

(0.80 - 0.60) x 12 880 = (2 576) A✓

Usage variance:
(AQ - SQ) x SP = (12 880 - 12 800) ✓x 0.60✓ = (48) A

Direct labour:

Price variance:
(AP-SP) x AH

Actual labour rate = R23 660/845 = R28

(28 - 25) x 845 = (2535) A✓

Efficiency variance:
(AH - SH) x SP

(845- 853.33) ✓ x 25 ✓= 208.25 F or 208.333F

Variable overheads:
Expenditure variance:
Budgeted Flexed variable overheads (AH X SR) – Actual Costs
(AH X SR) = 845 x 7.50 = 6 338 (rounded)
6 338 – 5 070 (Actual) = 1 268F✓

Efficiency variance:
(SH - AH) x SP
(853.33 - 845) ✓ x 7.50 ✓= 62 F (rounded) **or** 62.5 F

Fixed overheads:

Expenditure variance:
Budgeted expenditure – Actual expenditure
Budgeted fixed overheads = 22 500
Actual Fixed overheads = (11 000+ 6 000 + 8 000) = 25 000
22 500 – 25 000 = (2 500) A✓
Allocated fixed overheads = 22 500/15 000= R1.50 per packet

Volume variance:
(Budgeted units – Actual units) x SP
(15 000-12 800) ✓x 1.50✓ = (3 300) A

Calculation of capacity and efficiency variance:
Budgeted fixed overheads = 22 500

Capacity variance: **3 487.50A** ✓
Labour capacity (15 000*4/60 = 1000
Standard rate per hour (22 500/1000 = 22.50)
Actual hours at standard rate per hour 19 012.50
(845 x 22.50)

Efficiency variance: **187.50F** ✓
Fixed overheads recovered (Allocated)
(SQ x SP for actual production) 19 200
(12 800 x 1.50)
Volume variance = 22 500 - 19 200 = 3 300 adverse

Total fixed overhead variance = (2 500) + (3 300) = (5 800) A

Calculation of actual profit

Sales (12 800 x 17)	**217 600(1/2)**
Cost of sales:	**(97 844)**

Raw materials: Peanuts (385kg x R66/kg)		**(25 410) (1/2)**
Raw materials: Raisins (280kg x R30/kg)		**(8 400) (1/2)**
Plastic packets: (12 880 x R0.80/packet)		**(10304) (1/2)**
Labour: (given) (845 hours x R28)		**(23660)(1/2)**
Variable overheads (given) (845 x 6)		**(5 070) (1/2)**
Fixed overheads		**(25000)(1/2)**
Actual profit	**119**	**756 (1/2)P**

(c)
The various variances can be explained as follows:

Note to students:
- When answering this type of question, where you have to explain reasons for variances, start by identifying how many variances are actually required. This will ensure that you do not waste time doing unnecessary variances.
- Once you have understood the required, check the mark allocation that, will guide you concerning the extent of your discussion. The test question required four variances (sales, peanuts, raisins and packets) for eight (8) marks, so you didn't have to go into detail with regards to your explanations. However, if the required had specified that you must provide as much detail as possible or the mark allocation was larger, then you would have go into more detail and address mix and yield (and capacity and efficiency variance where applicable).
-The starting point for actually answering all your variance explanations is to write down your variance as well as its direction, for example, the sales price variance you would start with -'25 600A'. This will ensure that the marker gives you marks principally even if your answer and your explanation is wrong. However, if you do not start by listing them, you will not be marked principally and you may end up losing many marks.

Sales variances:
The favourable sales price variance is due to a price increase of $2.00. ✓This may be due to the popularity of Air Woolies products✓ or in response to higher than expected costs as the company will want to maintain their current mark-up percentage.✓

The adverse sales volume variance is due to sales being lower than expected. This may be a result of price increases or order quantities not being as large as they were in previous months.✓

Max: 2

Raw materials:

Peanuts:
The adverse price variance can be attributed to fluctuations in the rand/ dollar exchange rate. This is the only reason for the price variance as the underlying peanut price was fixed. The ZWL dollar weakened against the US dollar.✓ The adverse usage variance is a result of a

higher proportion of peanuts being used per packet.✓ Peanuts made up 57,89% of packets produced compared to the 55% composition which was budgeted. There was also abnormal and unexpected wastage of peanuts during the production process possibly due to poor control.✓

Max: 2

Raisins:
The favourable price variance is due to an unexpected discount which was received from suppliers.✓

The favourable usage variance is a result of a higher proportion of peanuts being used instead of raisins. Raisins made up 42,11% of packets produced compared to the 45% composition which was budgeted. Higher quantities of peanuts were possibly used due to a shortage of raisins. ✓

Max: 2

Packets:
The adverse price variance is a result of packet pricing changes, possibly due to unforeseen price increases by suppliers. Increased demand for these packets may have prompted suppliers to increase packet prices.✓

The adverse usage variance is a result of wastage Changes in the quality of packaging may have resulted in more packets breaking or being rejected during production✓. Is Food for Flight using more than one supplier, with one supplier providing poor quality packaging?✓

Max: 2

Note to part 4 students: Discussion of reasons for variances should be specific to a particular variance and cannot be a general discussion in total.

Direct labour:
The adverse price variance can be attributed to increased wage rate. This may be a result of strikes or workers receiving increased wages after negotiations with trade unions.

The favourable direct labour efficiency variance can be attributed to learning curve factors and increased efficiency not being taken into account when standards were set.

Variable overheads:
The favourable variable overhead expenditure variance is a result of actual variable overheads being lower than those expected. Food for Flight budgeted for cost increases/ hikes which were higher than those actually incurred.

The favourable variable overhead efficiency variance is a result of labour hours being less than expected due to increased efficiency and improved scheduling/ planning.

Fixed overheads:
The adverse fixed overheads expenditure variance is a result of **budgeted** fixed overheads exceeding actual overheads. Fixed cost pricing relating to rental, depreciation and electricity were not as high as those expected by Food for Flight. There were price increases for fixed amounts payable relating to rental, depreciation and electricity. Even if the price of one fixed cost was lower than standard costs, other fixed cost price hikes outweigh the benefit of any favourable amounts.

(Need to discuss price increases/ changes relating to all fixed costs to earn mark)

The adverse fixed overhead volume variance is due to the fact that 12 800 packets were produced instead of the 15 000 that was budgeted. This resulted in a lower total fixed overhead expenditure being allocated to packets as a result of production.

The adverse fixed overhead capacity variance is a result of producing fewer packets than the 15 000 packets which make up the normal capacity which Food for Flight are able to produce through their operations.

Any other valid point.

Available: 12
Max: 8

(d)
The following additional factors and strategic considerations should be taken into account:

How will meals be transported to airlines in other parts of Africa.✓

Will increased order quantities and products resulting from expansion impact on employee's work ethic and their efficiency.✓

Consider whether any additional new clients can be obtained due to airlines which regularly land in various locations in Africa.✓

Consider competitor's actions regarding expansion in Africa and the impact of this on Air Woolies market share✓
The company should consider any impact their production process and other operations will have on the environment✓
Consider developing new airline food products in response to consumer demands in different countries.✓

What impact will different exchange rates have on future financial performance?✓ Has Food for Flight considered the use of derivatives?✓

Consider laws and regulations which will need to be adhered to in countries outside Zimbabwe.✓
Consider whether the company has the financial capacity to expand (1)
Consider the financial viability of the expansion into Africa (1)
Consider whether the company has the adequate knowledge and skills to thrive in the African market (1)
Any other valid point✓

Semester Exercise on Standard Costing – Individual Assignment

Due date - 20 September 2022

KK biltong (hereafter 'the company' and 'KK biltong') is a Victoria Falls based biltong manufacturer and retailer. The company is famous for its tasty biltong snack packets which are sold at lower prices than that of competitors.

KK biltong oversees the entire process of the manufacture of the biltong; they buy the meat from a reputable local company. The meat is then cut into long strips and salted with a secret recipe which gives the biltong its special and unique flavour. The spice mixture is rubbed into the meat by hand and the salted strips are then transferred to a suitable container for further curing.

Thereafter the meat is dried in a solar dryer for a few hours. Once dry, the biltong is sliced by hand and packed into individual 100 gram packets. This is the only size of this product that they sell. The biltong is then sold to retailers.

Salt and pepper are the principal ingredients used in BB's secret spice recipe, although other ingredients such as sugar, coriander, vinegar and Worcestershire sauce also give their biltong its famous taste

The financial manager of KK biltong has told you that the company is very pleased with the current state of affairs at the company. He informed you that actual sales for April 2017 were 2 000 packets more than the 20 000 packets that were expected to be sold. However, he cannot understand why actual results for April 2017 differed so much from the budget as he has maintained the same labour force and the same suppliers for the month of April with the same strategic outlook as in prior months. He approached you, as a management consultant to look into this, and presented you with the following facts and figures:

Budget – April 2017

	Notes	$
sales		560 000
Meat	1	
Spices	2	
Packaging	3	
Labour	4	
Overheads	5	85 000
Net Profit		112 200

1) One third of the meat weight entered into the biltong manufacturing process is lost due to the drying process (I.e. 3 kilograms of raw meat will make 2 kilograms of biltong). Raw meat is purchased from Flash Meat distributors for $65 per kilogram
2) One kilogram of biltong meat (after drying) should use 50 grams of the special spice mix. KK biltong purchases the spices from Patel Spices in pre-packaged containers which cost $140 per kilogram.
3) The branded packaging is manufactured and imported from a supplier in Botswana who has proved to be more reliable and less expensive than local Zimbabwean competitors. The packets cost 0.5 Botswana Pula each.
4) The company employs 25 direct labourers who are paid an hourly rate of $30. These workers slice and spice the meat, place the pieces in the solar dryer and when dry, package the biltong into individual packets. Management expect each worker to complete 5 packets of biltong an hour. Each worker should work 160 hours a month in order to meet normal production.
5) Overheads are comprised of a fixed and variable portion. Examples of overheads include electricity used by the solar dryer, insurance and administrative salaries. In the previous month direct labour hours totalled 3 600 and overheads were $83 000.

Fixed overheads are allocated on direct labour hours based on normal capacity. The April 2017 budget assumes normal capacity.

Actual – April 2017

- Although more packets were sold than anticipated, actual sales were $10 000 less than expected.
- During April 2017 management used a more efficient drying method. Using this method, the loss from drying only amounts to 20% of the meat entering the biltong-making process. Flash Meat distributors increased their prices to $67 per kilogram.
- 117kg spices were used during the month at total cost of $14 960.
- 1% of the packets used in the month were stolen from the company's warehouse and had to be re-ordered from the Botswanan supplier. The price per packet in Botswanan Pula remained the same for this order as for the previous order.
- Direct labourers were paid at the expected rate. Each worker worked for 160 hours as expected by management.
- Variable overheads amounted to $6 per direct labour hour.
- Fixed overheads were $68 000 for the month.
- Actual profit for the month was $123 017.

Additional information

- The expected average exchange rate for April 2017 was 1 Botswana Pula = 1.38 Zimbabwe dollar. The actual average rate was 1 Botswana Pula = 1.42 Zimbabwean dollar.
- The company uses a standard absorption costing system.
- There was no opening or closing stock of any kind during April 2017.

Information for Part D only

The only other product that management sell in addition to the biltong is chilli bites. 20% of all packets sold should be chilli bites which have a selling price of $36 per packet. Only 4 000 packets of chilli bites were actually sold during April 2017.

REQUIRED		
		Total
(a)	Calculate the number of packets of biltong that KK biltong will need to sell in order to break even.	(12)
(b)	Calculate the margin of safety percentage if the budgeted number of packets for April 2017 was sold.	(2)
(c)	Reconcile budgeted contribution to actual net profit, clearly showing all variances (except for sales mix and yield variances).	(27)
(d)	Using the information given for part (d), calculate in as much detail as possible, the sales volume variances for biltong and chilli bites (at a sales price level).	(6)
(e)	Give reasons for the biltong sales and packaging variances.	(4)
(f)	List the purposes and benefits of using standard costing.	(4)

(g)	Explain whether standard costing would be better suited in industries where non-repetitive activities are observed or in industries where process costing is used. Give reasons for your answer.	(2)
(h)	Explain to the financial manager of KK biltong the difference between calculating the meat price variance based on quantity used and on quantity purchased in situations where these quantities differ.	(3)
Total for assignment		**60**

ABC COSTING

In traditional absorption costing overheads are charged to products using a predetermined overhead recovery rate. This overhead absorption rate (OAR) is based upon volume of activity. A full unit cost is computed in order to satisfy financial accounting requirements. However, it is always stressed that full product costs, using financial accounting principles, are not suitable for decision-making purposes. Instead decisions should be based on a decision-relevant approach incorporating relevant/incremental cash flows. With this approach, decisions such as introducing new products and special pricing decisions should be based on a study of only those incremental revenues and expenses that will vary with respect to the particular decision. This approach requires that special studies be undertaken when the need arises. However, studies have shown that the majority of companies base their decision making upon full product cost.

In the late 1980s Cooper and Kaplan developed a more refined approach for assigning overheads to products and computing product cost. This new approach is called **activity based costing (ABC). It is claimed that ABC provides product-cost information that is useful for decision-making purposes**.

Traditional systems measure accurately volume-related resources that are consumed in proportion to the number of units produced of the individual products. Such resources include direct labour, materials, energy and machine-related costs. However, many organisational resources exist for activities that are unrelated to physical volume. Non-volume related activities consist of support activities such as materials handling, material procurement, set-ups, production scheduling and first-item inspection activities. Traditional product-cost systems, which assume that products consume *all* activities in proportion to their production volumes, thus report distorted product costs.

The activity-based costing procedure

Cooper and Kaplan stated that it was the support activities that were the cause of many overheads, for example, material handling, quality inspection, setting up machinery, material acquisition, etc. Thus a simple three-step philosophy was developed:

- support activities cause cost;
- the products consume these activities;
- cost should, therefore, be charged on the basis of consumption of the activities.

Method

1 Identify the organisation's major activities.

Ideally about 30 to 50 activities should be identified. However, over time, some large firms have been known to develop hundreds of activities. A suitable rule of thumb is to apply the 80/20 rule: identify the 20% of activities that generate 80% of the overheads, and analyse these in detail.

2 Estimate the costs associated with performing each activity – these costs are collected into **cost pools**.

3 Identify the factors that influence the cost pools. These are known as the cost drivers. For example, the number of set-ups will influence the cost of setting up machinery.

4 Calculate a cost driver rate, for example a rate per set-up, or a rate per material requisition, or a rate per inspection**. Cost driver rate = Cost pool/ Level of cost drivers**

5 Charge the overheads to the products by applying the cost driver rates to the activity usage of the products.

To appreciate the application of this procedure it is necessary to work through an example.

FAVOURABLE CONDITIONS FOR ABC

The purpose of moving from a traditional costing system to an activity-based system should be based on the premise that the new information provided will lead to action that will increase the overall profitability of the business.

This is most likely to occur when the analysis provided under the ABC system differs significantly from that which was provided under the traditional system, which is most likely to occur under the following conditions:

- when production overheads are high relative to direct costs, particularly direct labour;
- where there is great diversity in the product range;
- where there is considerable diversity of overhead resource input to products;
- when consumption of overhead resources is not driven primarily by volume.

Information from an ABC analysis may indicate opportunities to increase profitability in a variety of ways, many of which are long-term. For example, an activity-based analysis may reveal that small-batch items are relatively expensive to produce, and are therefore unprofitable at current prices.

A number of responses to this information could be adopted. The first response might be to consider stopping production of such items, and concentrate on the apparently more profitable high-volume lines. Another approach would be to investigate how the production process could be organised in such a way as to bring the cost of producing small-batch items closer to that of producing high-volume goods.

By identifying the cost of carrying out particular activities, the new approach provides opportunities for directing attention to matters of cost control. It can therefore be viewed as a much longer-term technique than the word ‘costing’ in the title suggests. The establishment of an ABC product cost may thus be considered to be the beginning of the process, rather than an end in itself. The recent use of the term *activity-based management* suggests this forward looking orientation, which is assuming increasing importance.

A summary of benefits and limitations

Benefits

1 Provides more accurate product-line costings particularly where non-volume-related overheads are significant and a diverse product line is manufactured.

2 Is flexible enough to analyse costs by cost objects other than products such as processes, areas of managerial responsibility and customers.

3 Provides a reliable indication of long-run variable product cost which is particularly relevant to managerial decision making at a strategic level.

4 Provides meaningful financial (periodic cost driver rates) and non-financial (periodic cost driver volumes) measures which are relevant for cost management and performance assessment at an operational level.

5 Aids identification and understanding of cost behaviour and thus has the potential to improve cost estimation.

6 Provides a more logical, acceptable and comprehensible basis for costing work.

Limitations

1 Little evidence to date that ABC improves corporate profitability.

2 ABC information is historic and internally orientated and therefore lacks direct relevance for future strategic decisions.

3 Practical problems such as cost driver selection.

4 Its novelty is questionable. It may be viewed as simply a rigorous application of conventional costing procedures.

Practice Question 1

Gempatch Ltd manufactures three products using the same plant and processes. The following is relevant to the September 2000 production period:

	Product		
	Jaspis	**Agate**	**Sodalite**
Volume	5 500	600	7 000
Material cost per unit	$50	$160	$170
Direct labour per unit	1/2 hour	2 hours	1 & 1/2 hour
Machine time per unit	1/4 hour	1 hours	1 & 1/2 hour
Labour cost per unit	$30	$120	$90

The manufacturing overheads for the period can be analysed as follows:

	$000
Machine activity	350
Set - up costs	44
Ordering costs	20
Material handling	75
Spare part administration	10
	499

Currently the overheads are absorbed by the products based on machine hours at a rate of $40 per machine hour, which gives the following overheads per product:

Jaspis	$10
Agate	$40
Sodalite	$60

The company is at present considering a change to activity based costing. An investigation into the manufacturing overheads activities for the period revealed the following:

	Total	Product Jaspis	Agate	Sodalite
Number of set - ups	16	6	2	8
Number of orders	10	5	1	4
Number of material handlings	27	12	3	12
Number of spare parts	12	7	1	4

REQUIRED

(a) Contrast the features of a business that will benefit from activity based costing (ABC) with those of a business that will not. (4)
(b) Determine the overheads per product unit for Gempatch Ltd by using ABC and briefly comment on the differences in the overheads per unit determined with the current system and the ABC system. Your calculations must be rounded off to two decimals. (19)

Practice Question 2

Question 2 – Activity Based Costing (32 marks) – Unisa Adopted

Sweet Roast Limited is a producer of a variety of high grade coffee blends. The firm buys coffee beans mainly from the Ethiopian Highlands. The coffee beans are then roasted, blended and packaged in one kilogram bags for sale to specialist coffee houses around the world. The major raw material in the coffee blends is the beans. The company's automated roasting, blending and packaging processes result in substantial overheads. Since the company is highly automated it uses very little direct labour.

Coffee blends from the mainly Ethiopian Highlands are very popular and sell in large volumes while blends from Tanzania and Malawi sell in moderate volumes. The company has a mixed pricing strategy.

The company's budget for the coming year includes estimated production overheads of $2 200 000. The company assigns production overheads to products on the basis of direct labour hours. The expected direct labour cost is $600 000 and the budgeted directed labour hours are 50 000. Based on the sales budget and the material requirement budget, the company will, during the forthcoming year, purchase and use $5 000 000 of raw materials, mostly in the form of coffee beans.

The budgeted costs for direct materials and direct labour for one-kilogram bags for two of the company's coffee products are as follows:

	Ethiopian Roast	Malawi Roast
	$	$
Direct materials (per 1 kg bag)	4,50	2,90
Direct labour (per 1 kg bag) (0,02 Hours per bag)	0,24	0,24

The company's marketing manager is of the belief that the company's traditional costing system may be providing misleading cost information related to the products. She has learnt of alternative methods of assigning manufacturing overheads to products and has collected the following information for the **total production** in the company:

Cost pool	**Measure**	**Yearly Activity of all products**	**Budgeted Annual Cost**
			$
Purchasing	Purchase orders	2 000 Orders	560 000
Material Handling Costs	Number of set-ups	1 000 Setups	193 000
Quality control	Number of batches	500 batches	90 000
Roasting	Roasting hours	95 000 hours	1 045 000
Blending	Blending hours	32 000 hours	192 000
Packaging	Packaging hours	24 000 hours	120 000
Total manufacturing overhead			2 200 000

Data regarding the expected production for **the two selected products** is given below:

	Ethiopian Roast	**Malawi Roast**
Forecast sales and production quantities)	80 000 kg	4 000 kg
Batch size	5 000kg	500 kg
Setups	2 per batch	2 per batch
Purchase order size	20 000 kg	500 kg
Roasting, blending and packaging time per 100 kg	2,3 hours	2,3 hours

The company holds no inventory of unroasted coffee beans at either the beginning or the end of the financial year.

REQUIRED		**Marks**
(a)	Using direct labour hours as the base for assigning manufacturing overhead cost to products, undertake the following: (i) Calculate the predetermined overhead recovery rate that will be used during the forthcoming year. (ii) Calculate the unit product cost of one kilogram each of the **Ethiopian and Malawi roast** products.	(1) (3)
(b)	Using activity based costing as a basis for assigning manufacturing overhead cost to products, undertake the following: (i) Calculate the total amount of manufacturing overhead cost that should be assigned to the total production of each of the roast blends for the year. (ii) Using the data calculated under b(i) above, compute the overhead cost per kilogram of each of the Ethiopian and Malawi roasts. (Work to 2 decimal places) (iii) Calculate the unit product cost of one kilogram each of the Ethiopian and Malawi roast products	(19) (3) (3)
(c)	In bullet point form, prepare short notes for the marketing manager on your findings under (a) and (b) above and clearly indicate the implications to the company of using direct labour as the base for assigning manufacturing overheads to products.	(5)

Practice Question 3 – UNISA Adopted

Mr Terry, believes that the traditional costing system may be providing misleading information. He has developed an activity based analysis of the 2010 manufacturing overhead costs which is shown in the following table:

Activity	**Cost Driver**	**Cost**
		R'000
Materials handling	Set-ups	21 300
Quality Control	Batches	2 100
Baking	Baking hours	10 500
Packaging time	Packaging hours	4 600
Not identifiable	-	5 500

The actual data regarding the 2010 production of rusks are presented below:

	Buttermilk	**Muesli**
Batch size	10 000 boxes	10 000 boxes
Set-ups	3 per batch	4 per batch
Baking time	1 hour / 100 boxes	1 hour / 100 boxes
Packaging time	0,1 hour / 100 boxes	0,1 hour / 100 boxes

REQUIRED		**Marks**
(a)	Using an activity based approach, calculate the actual manufacturing overhead cost of buttermilk and muesli rusks for the year ended 28 February 2010.	(12)
(b)	Prepare a memo to Mr Terry to briefly explain the impact of the ABC approach on the allocation of the two products' manufacturing overhead cost.	(4)
(c)	Calculate Lekker Rusks (Pty) Ltd's Rand budgeted break-even point based on expected sales of 1 350 000 boxes buttermilk and 450 000 boxes muesli.	(4)
	Total	20

DECISION MAKING

The management accountant's role in decision-making is to provide information relevant to a decision. Managers use this information in preparing a quantitative analysis of the problem under review. Qualitative factors should also be considered before a final decision is taken.

The various decisions that are discussed in the textbook, includes the following:

- Limiting factor decisions
- outsourcing versus manufacturing (example 9.4)
- deletion of a market segment
- expansion of a product range
- acceptance of once-off sales opportunities at discounted selling prices (example 9.5)
- further processing of a product (examples 9.6 & 9.7)
- acceptance of special orders requiring the use of optimisation (see chapter 8).

Relevant Cost and Revenues

Decision making involves making a choice between two or more alternatives. The decision will be 'rational'; profit maximising. All decisions will be made using *relevant* costs and revenues.

'Relevant costs are future cash flows arising as a direct consequence of the decision under consideration.'

There are three elements here:

Cash flows. To evaluate a decision actual cash flows should be considered. Non-cash items such as depreciation and inter-divisional charges should be ignored.

Future costs and revenues. This means that past costs and revenues are only useful insofar as *they* provide a guide to the future. Costs already spent, known as *sunk costs,* are irrelevant for decision making.

Differential costs and revenues. Only those costs and revenues that alter as a result of a decision are relevant. Where factors are common to all the alternatives being considered they can be ignored; only the differences are relevant.

In many short-run situations the fixed costs remain constant for each of the alternatives being considered and thus the marginal costing approach showing *sales, marginal cost* and *contribution* is particularly appropriate.

In the long run (and sometimes in the short run) fixed costs do change and accordingly the differential costs must include any changes in the amount of fixed costs.

Sunk cost: The cost of resources already acquired where the total cost will be unaffected by the choice between various alternatives. These costs have already arisen and are as a result of a past decision. In addition, they will not be changed by a decision made in the future. Sunk costs are irrelevant for decision making because they will remain unchanged irrespective of which alternative is chosen.

Committed cost: Costs that are subject to a binding contract in the future and are therefore not relevant to the decision.

Opportunity Cost

Opportunity cost is an important concept for decision-making purposes. It is the value of the best alternative that is foregone when a particular course of action is undertaken. It emphasises that decisions are concerned with choices and that by choosing one plan there may well be sacrifices elsewhere in the business.

Example 1

A company which manufactures and sells one single product is currently operating at 85% of full capacity, producing 102,000 units per month. The current total monthly costs of production amount to £330,000, of which £75,000 are fixed and are expected to remain unchanged for all levels of activity up to full capacity.

A new potential customer has expressed interest in taking regular monthly delivery of 12,000 units at a price of £2.80 per unit.

All existing production is sold each month at a price of £3.25 per unit. If the new business is accepted, existing sales are expected to fall by 2 units for every 15 units sold to the new customer.

What is the overall increase in monthly profit which would result from accepting the new business?

A £1,200

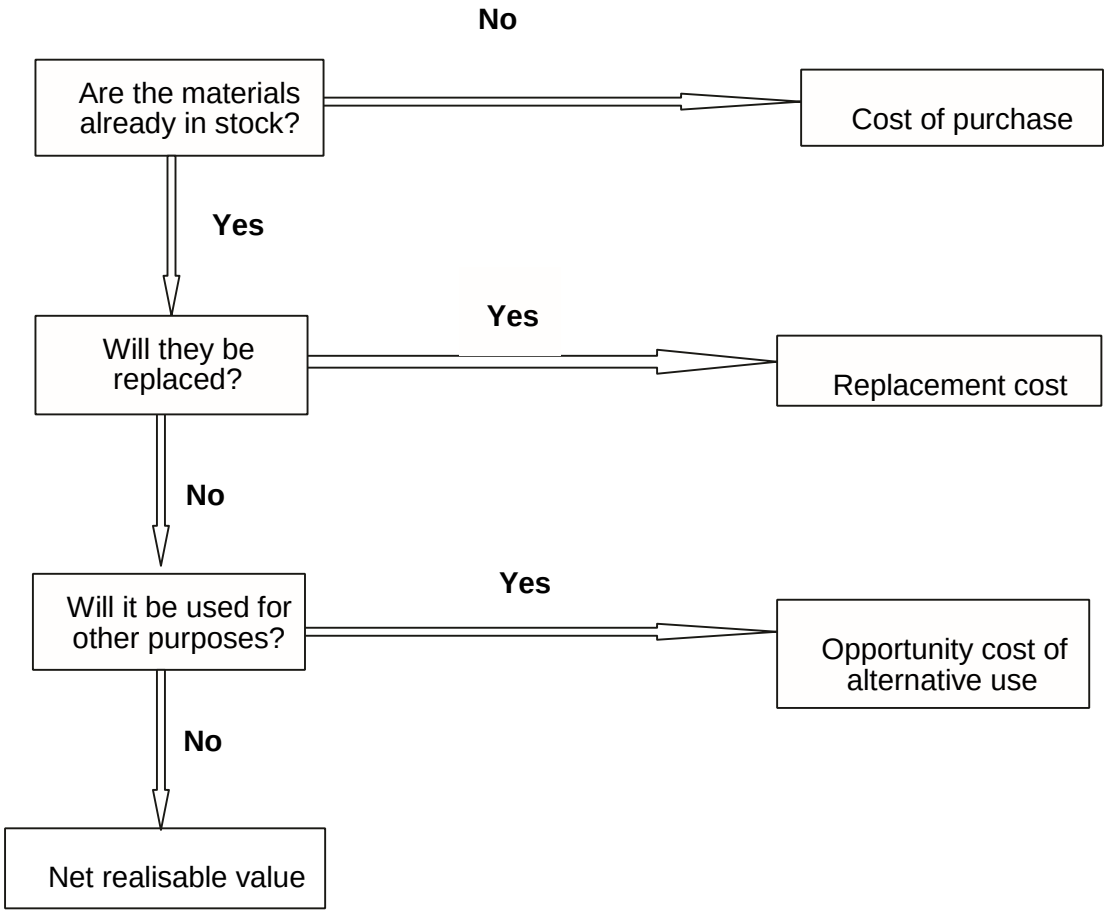
B £2,400

C £3,000

D £3,600

The Relevant Cost of materials

If a particular material or component is needed for a course of action the relevant cost of that material must be considered carefully. The following decision model may be used:



The flow chart hopefully provides a useful reminder of the points that must be considered when ascertaining the relevant cost of materials.

Do remember to apply basic logic, consider relevant cash flows, and then use a bit of common sense!

EXAMPLE 2 – RELEVANT COST OF MATERIAL

A firm is considering undertaking a contract for a special client.

The contract requires 200 kg of material D. There is a stock of 1,000 kg, which was bought for £10/kg last year. The current resale value is £5/kg, although to buy in any extra kilograms would cost £17/kg. There is no other use for material D, except as a substitute for material E, which is in constant use. Details of material E are shown below:

	£/kg
Original purchase price	8
Resale value	3
Replacement cost	15

What is the relevant cost of material D?

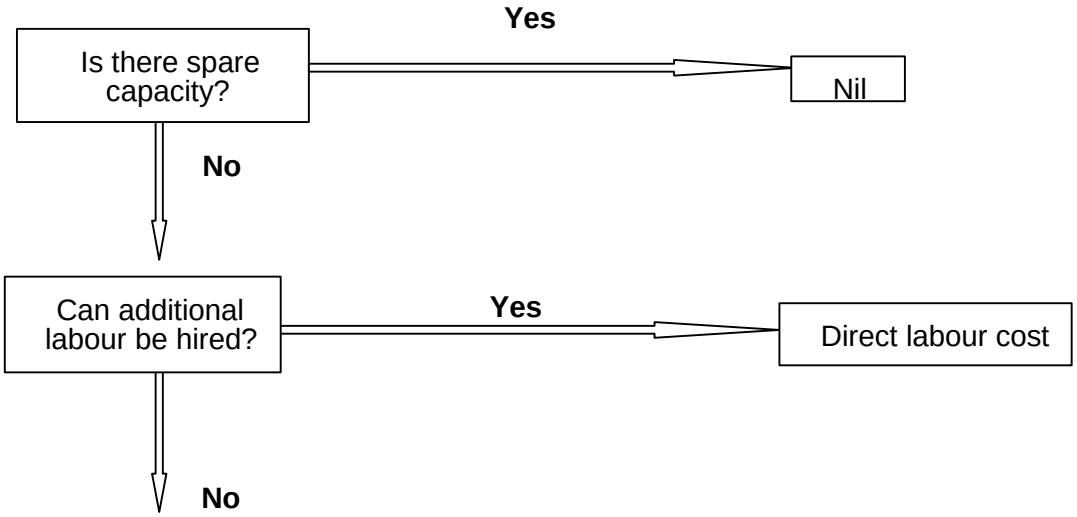
A nil

B £1,000

C £2,000

D £3,000

THE RELEVANT COST OF LABOUR



Contribution foregone plus direct labour cost

Again, remember to apply basic logic, consider relevant cash flows, and then use a bit of common sense!

EXAMPLE 3 – RELEVANT COST OF LABOUR

100 hours of skilled labour are needed for a special contract. The staff are working at full capacity at the moment and the workers would have to be taken off production of a different product in order to work on the special contract. The details of the other product are shown below:

	£/unit
Selling price	60
Direct material	10
Direct labour 1 hour @ £10/hour	10
Variable overheads	15
Fixed overheads	15

The skilled workers' pay-rate would not change, regardless of which product they worked on. What would be the relevant cost? **A** nil **B** £1,000 **C** £2,500 **D** £3,500

Long- term decision making

The approach to answering a short-term or long-term type question is very different. Therefore, it is very important that you identify within a scenario whether it is requiring a short run or long run decision to be made. For short run decisions, the understanding and application of the following concepts is very important: relevant costs, sunk costs, opportunity costs/income, incremental costs, avoidable costs etc.

Where a long run decision is required, students must be able to use the information in the question to evaluate whether the income from the project covers all the costs in the longterm. Usually a net present value is required in order to do this. In the long run, most costs and revenues are relevant, as they can be changed by a decision, however in the short-term they may not be.

Comprehensive Example – Relevant Costing

The manager of VO Ltd has been approached by KZC Ltd to provide a quotation for the installation of a **Fibre optic uptake** for their Company Internet. The inexperienced manager of VO responded that the quote would be done on a relevant costing basis. As the management accountant of VO ltd you have been provided with the following information.

1.The manager of VO
- Took four days to cost the requirements; this took place because of the potential order.
- Cost per day is $3 500.

2. The project would require 2 600 metres of 30 Mbps cabling, which is in regular use. There are 1 200 metres in inventory at a cost of $9 350 per metre with the current cost being $9 700 per metre.

3. 950 metres of 50 Mbps would be required for the project. This is not regularly used and would have to be purchased specifically for the project. The supplier with the best pricing supplies in quantities of 1 000 m @ $12 600 / metre. The closest other price is $12 985 / metre for the specific length required.

4. 300 metres of 100 Mbps would be required at a cost of $16 400 / metre. VO has 200 metres in inventory at $15 900 / metre which was provisionally booked out as a sale at $16 000/metre as it was not anticipated to be used again. The client involved would accept $50 000 as a cancellation fee.

5. 600 complete connectivity units. VO obtains these from a group company at $7 980 each, on which the group company makes a mark-up of 40% to cover fixed costs. They currently have capacity to produce and deliver 370 units on demand. With effect from yesterday the group CEO posted the following notice "It is now group policy to value all internal transfers of

inventory at Variable cost". The Connectivity units can also be obtained externally at a market price of $7 980.
6. A digger machine is required for the project. VO does not have spare capacity on the required digger machine. This machine can either be hired on a monthly basis at $100 000 for a month or on a daily basis at $6 000 per day. A minimum of 12 days' work will be required for the project.
7. 3 300 man-hours of labour will be required at a rate of $100 / hour. VO has spare capacity of 2 400 hours from existing paid workers. Workers can either be redeployed from other projects which deliver contribution of $40 per labour hour or be paid overtime at normal industry overtime rates of 50%.
8. 100 hours of high-skilled engineering supervision will be required at $750/hour, also the going market rate. A normal work week entails 40 hours should an external engineer have to be hired. One of VO's engineers offered to work 25 hours overtime on the project without claiming for the overtime.
9. A fixed overhead allocation of $2 million is made on all projects of this size and scope.

Required:

(a) Prepare a schedule of all relevant costs for the fibre optic project from the VO perspective. Inclusion or exclusion of costs should be motivated.
[Round all amounts to the nearest $'000] (**12 marks)**

(b) Provide a brief explanation for the term Differential Costs **(1 mark)**

COMMON TYPES OF DECISION

1 Limiting factor decisions

2 Make or buy decisions

3 Shutdown decisions (deleting a segment)

4 Accept or reject an order decisions

5 Pricing decisions

6 Joint product and further processing decisions

Limiting factor

Also known as principal budget factor or key budget factor. It is that factor which prevents a company from achieving the output and sales that it would like. In practice the limiting factor is often sales demand, but in the exam it may be a shortage of labour in total or a shortage of one particular grade of labour or a shortage of material or machine capacity or capital or anything the Examiner fancies.

Single limiting factor

If there is a just one limiting factor then the rule is to maximise the *contribution per unit of scarce resource*. The contribution per unit of each product is calculated and divided by the amount of scarce resource each product uses. The higher the contribution per unit of scarce resource the greater the priority that should be given to the product. Once the priorities have been decided the scarce resource is allocated to the products in the order of the priorities until used up.

Example 3:

Sable Chemicals Ltd produces three chemicals X, Y and Z. The selling price and cost per litre for each of these products are budgeted as follows:

	X **$/Litre**	**Y** **$/LITRE**	**Z** **$/Litre**
Selling price	100	120	120
Direct materials	20	16	21
Direct labour ($6 per direct labour hour)	18	24	27
Other direct expenses	-	3	-
Variable overhead	12	16	18
Fixed overhead	6	8	9

Notes:

1. The fixed overhead is absorbed on the basis of labour hours, based on a budget of 440 hours per month.

2. Maximum demand for each product for month 4 is as follows:

   X = 150 litres

   Y = 40 litres

   Z = 60 litres

3. Included in the maximum demand totals is an unavoidable commitment to a major customer to supply 15 litres per month of each of the three products.

4. During month 4 there is a shortage of labour hours that will restrict production. The total number of labour hours available is 375 hours.

5. Sable is able to produce and sell fractions of a litre.

Required		**Marks**
(a)	Determine the production mix that will maximise profit in month 4 and calculate the resulting profit.	12
(b)	After completing the production plan you are informed that new environmental controls on pollution are to be introduced from the beginning of month 4. These controls relate to the production of product Z only, and will incur an additional fixed cost of $1 000	4

	per month for each month that Z is manufactured. In addition you also learnt that an overseas supplier will supply as many litres of chemical Z as Sable Ltd needs at a cost of $100 per litre. Importation of Z will not incur the additional fixed pollution control costs**. Using the above information calculate how much chemical Z (if any) should be purchased by Sable Ltd from the overseas supplier in month 4.**	
(c)	Briefly explain FOUR factors, other than costs or selling price, which should be taken into consideration when deciding whether to subcontract the manufacture of chemical Z.	4

Solution

Practice Questions

Question 1 – Adopted from NUST past question paper CAC 4105 -2021

QUESTION 4 – APPLICATION OF MARGINAL COSTING PRINCIPLES.

The Floor Cleaner Division manufactures three types of floor cleaner, namely Product F, Product L and Product O, which are sold in 750ml bottles. Details of the products are as follows:

Product:	F $/bottle	L $/bottle	O $/bottle
Selling price	30,00	22,00	26,50
Liquid A ($16 per litre)	4,80	3,20	5,60
Liquid B ($14 per litre)	4,20	3,50	2,80
Other ingredients and packaging	5,00	3,50	4,00
Variable overhead ($140 per machine hour)	4,20	2,80	4,20
Fixed overhead ($160 per machine hour)	4,80	1,60	1,60
Manufacturing profit	7,00	7,40	8,30

The machines used in this process can only be utilised for this particular manufacturing process. The demand for the three products for February 2022 has been estimated as follows (excluding fixed contracts):

	F	L	O
Bottles	1 000	2 100	1 500

In addition to the demand above, fixed contracts with lodge clients have been signed for the following quantities for February 2022:

	F	L	O
Bottles	400	500	460

The Floor Cleaner Division does not hold any inventory of raw materials or finished goods.

The management of BBCP has advised that the availability of Liquid A and Liquid B will be restricted during February 2022 to 1 800 litres of Liquid A and 1 200 litres of Liquid B, as advised by the only local supplier of these liquids. BBCP has no intention

of venturing into the import market for these liquids. No restrictions are expected on any other resources.

QUESTION 4			
REQUIRED		**Marks**	
		Sub-total	**Total**
(a)	Calculate the optimum product mix to maximise BBCP's floor cleaner division's profit for February 2022. Also calculate the number of bottles of product that cannot be supplied due to the shortage of Liquid A and/or Liquid B.	(18)	(18)
(b)	Briefly list BBCP's major risks in terms of liquids A and B and then explain why, in your opinion, it is reasonable for BBCP to import these liquids. [Risks: 3 marks, import acquisition 4 marks]	(7)	(7)
Total for Question 4			**25**

Examiner Comments *– the question was poorly and disappointingly done with a lot of students exhibiting a very poor foundation or background of management accounting techniques. Some students failed to interpret the requirements of the question and could not tell the question required them to apply Limiting factor concepts.*

Question 2 – Adopted from NUST past question paper CAC 4105 -2021

QUESTION 2 – DECISION MAKING PART A - relevant costing

The Bulawayo City Council is hosting a cultural day for schools in the City at ZITF Grounds and has asked Hotdog to prepare a quote to provide each child with a food package consisting of: One Easy-dog Hotdog, One Soft drink, One Health bar. For the tender, the MD will assume a price for the Easy-dog of $12,50 from the Supply Division. The MD is eager to get the contract as there are many of these events country wide and he had a few ideas to maximise his profit:

- 350 children are expected to attend. As Hotdog wants to promote its brand, it said it will provide 10% additional packages at cost to cover for any additional children attending as well as free delivery.
- Soft drinks will have to be acquired. The current market price is $5,50 each. There are currently 150 cans in inventory which was acquired at $5,20 to send to the franchise-holders.

- Hotdog plans to provide an all-in-one parcel to the children by packing everything into a polystyrene box. Since this differs from their normal inventory, it will have to be acquired from a new supplier. There is however a minimum order quantity of 400 at 50c per box
- The health bars are also not sold at the stores and will have to be acquired at $4,50 each. The minimum order quantity is 2 000 bars. The MD suggested that they sell the rest of the bars at their Bulawayo stalls as an experiment.
- For packing and quality control of the parcel, workers will have to work overtime. The MD will have to work 20 additional hours to prepare the tender. His currently hourly rate is $450. Three wage earners who normally earn $35/hour will work 4 hours' overtime each for two days. The overtime rate is calculated at 1,5 times the normal wage rate. The Supply division are expected to work 10 hours' overtime in preparation of the Easy-dogs. These employees earn the same wage rates as the other wage labourers. Hotdog subscribes to the Basic Conditions of Labour/Employment act.
- The MD has found that this will be the ideal time to buy a new delivery vehicle he was planning on for a while now to help with operations at a cost of $285 000. ZIMRA has allowed an annual 20% wear and tear deduction.
- Fixed overhead is allocated at a rate of $20 per labour hour.
- Variable overhead is expected at $2,50 per parcel. The MD expects a 20% mark-up on relevant cost.

QUESTION 2			
REQUIRED		**Marks**	
		Sub-total	**Total**
(a)	Calculate the expected selling price per food package for the special order linked to the Cultural Day on a relevant costing basis. Clearly show the relevant cost value for each of the items. Explain each relevant value you have evaluated and why the values you have excluded are not relevant.	(8)	
	Discuss whether it is likely that the municipality will accept the deal.	(2)	(10)

Practice Question 3

SHEPS Ltd is involved in the design and manufacture of train engines. The company has just received an enquiry about the possibility of supplying 30 engines to a privatised transport company NRAZ. The finance director of NRAZ has informed SHEPS Ltd that they have just received quotes from other manufactures of train engines and that the most reasonable quote was $145 000 per engine. The management accountant has provided you with the following details relating to the costs involved in the construction of the train engines:

1. Each engine will require 12 litres of oil. The company has 300 litres of oil in stock and if the oil is not used it will be disposed immediately. The cost of disposal is $1, 850. The original purchase price of the oil in stock was $340 per litre. The replacement cost is $350 per litre of oil.
2. Each engine will also require 15 sheets of alloy steel. The purchase price of a sheet of steel is $3, 800 for all purchases up to and including 300 sheets. The supplier has agreed to offer a discount of 10% on the purchase price on all purchases over 300 sheets.
3. Additionally, each engine will require 1 slab of cast iron and this type of iron is used regularly by SHEPS Ltd. The company holds 22 of these slabs in the warehouse at present. These slabs of cast iron cost $3,415 per slab.
4. The construction of the engines will require a combination of skilled and unskilled labour. Each engine will require 35 skilled labour hours and 10 unskilled labour hours. The skilled labourers are paid $180 per hour and the unskilled labourers are paid 55% of the skilled labour hourly rate. If this contract does not go ahead there will be 300 skilled labour idle hours and he company is reluctant to make redundancies due to continued loyalty towards its staff. The unskilled labour will have to be hired in for the contract.
5. The project will require a project manager to oversee the work. SHEPS Ltd currently employs a manager with the necessary experience who will be transferred to the proposed project should it go ahead. This manager currently earns $78,000 per annum. Due to the size of this project SHEPS Ltd has agreed to pay him an additionally 5% of his current salary. The project manager will then be replaced by a less experienced manager who will be paid an annual salary of $65,000.
6. Variable overheads are absorbed at rate of $102 per skilled labour hour.
7. Fixed overheads will increase from their current level of $1,360,000 to $2,175,000 if the project to produce engines is undertaken.
8. In order to assess the practicalities of taking on this project SHEPS Ltd employed a company to research the availability of materials and experienced labour. This research cost $11,000 and $4,000 is still outstanding. If SHEPS takes on the project future research into safety issues will be required at a cost of $5,500.

Required:

(a) Provide a brief explanation for each of the following terms:
   - **i. Sunk costs**
   - **ii. Committed costs**
   - **iii. Incremental cost**
   - **iv. Relevant cost**

(b) Using relevant costing principles, determine whether or not SHEPS Ltd should undertake the contract. Your answer must include an explanation for the inclusion for the inclusion or exclusion of each of the points from points 1 to 8 above.

(c) List

METHODS OF COST DETERMINATION (DIRECT AND ABSORPTION COSTING)

The most important objective of product costing is to allocate the total manufacturing cost incurred during a certain period to the total number of units manufactured during that period, in order to calculate the manufacturing cost per unit. The manufacturing cost per unit is therefore calculated simply by dividing the total manufacturing cost for a certain period by the number of units manufactured during the period in question,

When the unit cost is known, this makes it easier to calculate the allocation of manufacturing cost to units that are still in stock and to units that have been sold. This information is required to calculate the net profit of the enterprise for the period, which is one of the most important goals of cost accounting.

There are two divergent schools of thought on what should be included in unit cost. Some people prefer absorption or full costing, but others favour direct or variable costing (also known as MARGINAL COSTING).

The concept of absorption costing is taken to include both fixed and variable cost as integral parts of the total manufacturing cost of a product. The concept of direct costing, on the other hand, includes only the variable cost in the manufacturing cost of a product.

THE USES OF DIRECT AND ABSORPTION COSTING

Direct costing is also known as variable costing or marginal costing. Only variable manufacturing cost, namely direct materials, direct labour and variable manufacturing overheads, are taken into account when manufacturing cost is calculated in total or per unit. All variable costs are taken into account, however, when marginal income is calculated. Fixed manufacturing overheads are written off against income as period cost for the period in which they are incurred.

According to the absorption costing method fixed manufacturing cost are included in the cost of a product. The product therefore “absorbs” both variable and fixed manufacturing cost. The larger the number of units of a product which are manufactured the lower the unit cost because the same amount of fixed cost is divided among a larger number of products. When absorption costing is applied, fixed cost are recovered on the basis of the number of units of a product manufactured during the period.

When the number of units produced and the number of units sold are the same, that is when there is no stock on hand, income statements drafted according to direct and absorption costing methods will show the same net profit. If there is stock on hand, the difference in net profit shown in income statements drafted according to the two methods can be reconciled by taking into account the difference between opening and closing stock according to the two approaches.

Information **casg**

Practice Questions

Question 1– UNISA Adopted

Lekker Rusks (Pvt) Ltd is a processor and distributor of the old South African favourite, rusks. The owner Mr Biscuit, uses his grandmother’s secret family recipe to manufacture and package rusks for resale.

Lekker Rusks currently offers two types of rusks, buttermilk and muesli, both which are processed and sold in 1kg boxes.

Budgeted Data for the 28 February 2010, year-end are as follows:

		Buttermilk	**Muesli**	**Company**
Units sold(boxes)		1 500 000	700 000	
Per Unit (dollars)		**$**	**$**	**$**
Sales price		38,5	55	
Direct materials		12	15,5	
Cost in Total (Dollars)				
Manufacturing Overheads	Fixed			45 000 000
Manufacturing Labour	Fixed			25 000 000
Admin Salaries	Fixed			777 000
Sales and Distribution	Fixed and variable			8 280 000

Mr Kuda, the management accountant, provided the following information in respect of the actual figures for the year ended 28 February 2010:

Sales	$/unit	39	57	
Materials	$/unit	12,7	15	
Manufacturing Overheads	$			44 000 000
Manufacturing Labour	$			24 000 000
Admin Salaries	$			777 000
Sales and distribution	$			8 000 000
Inventory - 1 March 09 ($)		2 000 000	1 485 000	
Inventory - 28 February 2010		?	?	

No inventory of raw materials or work in process are held.

All baked rusks are packaged and taken into inventory.

Assume that there are no expired goods on hand. i.e. all inventory have been sold before expiry date and thus nothing should be written off.

1. Assume that the budgeted closing inventory = actual closing inventory (per above quantity schedule).

2. The inventory on 1 March 2009 includes fixed manufacturing overhead and fixed manufacturing labour.

3. Manufacturing labour and fixed overheads are based on normal capacity of 1 000 000 hours per year.

For the year ending 28 February 2010 the following information is applicable:

	Buttermilk	**Muesli**
Budgeted labour hours	450 000	550 000
Actual labour hours	460 000	555 000

4. Admin salaries and sales and distribution costs are divided on a 50/50 basis between the two products.

$3 000 000 of the budgeted sales and distribution costs are fixed and the actual fixed costs equalled the budgeted fixed costs.

5. There is no difference between budgeted and actual depreciation.

REQUIRED		**Marks**
(a)	Prepare an income statement for the year ended 28 February 2010 on the absorption costing basis. **Show as much detail as possible and round all calculations off to 3 decimals.**	(20)
(b)	Calculate Lekker Rusks (Pty) Ltd's Rand budgeted break-even point based on expected sales of 1 350 000 boxes buttermilk and 450 000 boxes muesli.	(5)
	Total	25